

UBT008: CELL BIOLOGY AND GENETICS

L	T	P	Cr
3	0	0	3.0

Course Objective: The course is aimed to impart knowledge of structural and functional aspects of cells as unit of living systems. To understand functions of various organelles and transport of information and matter across cell membrane and classical genetics comprising Mendelian laws of inheritance and their significance in genetic diseases.

Syllabus

Cell structure and function: Cell – structural and functional unit of life, cell morphology, difference between bacterial, plant and animal cells, structure and function of cellular organelles, plasma membrane and cell wall, cytoskeleton, transport across cell membrane.

Cell division: Mitosis and meiosis, cell cycle and its regulation, apoptosis and necrosis.

Genetics: Mendel's laws of inheritance, Intra-allelic and inter-allelic interaction, incomplete dominance, codominance and blood types, Multiple alleles, Lethal genes, Polygenic inheritance, cytoplasmic inheritance, chromosomal structure, nucleosome and chromatin, sex determination and sex-linked inheritance.

Linkage and recombination: Recombination and crossing over, linkage maps, mapping of genes, Hardy-Weinberg distribution.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Acquire knowledge about the organizational and functional aspects of cell and organelles.
2. Learn about the interactions of the cells with outside environment through exchange of information and transport of molecules.
3. Learn about the classical genetics and transmission of characters from one generation to the next which will make foundation for the advanced genetics.
4. Develop innovative research ideas for curing genetic disorders in humans.

Text Books

1. Bruce Alberts et al., Essential cell biology, Garland Science (Taylor & Francis Group).
2. Gardner, Simmons and Snustad, Principles of Genetics by John Wiley & Sons.
3. Veer Bala Rastogi, Cell Biology (MedTech Science Press).
4. Eddon John Gardner. Principles of Genetics (Wiley; 8th edition).

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Reference Books

1. H Lodish et al., Molecular Cell Biology (4th edition), WH Freeman.
2. MW Stickberger, Genetics (3rd edition), PHI Learning Pvt Ltd (2012).
3. Geoffrey M. Cooper., The Cell: A Molecular Approach (Sinauer Associates Inc).
4. Peter J. Russel. *i*Genetics: A Molecular Approach (Pearson Education India).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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